Sheet 1

States and Gibbs operators

1. **Thermal spin.** For $H = (\hbar\nu/2)\sigma_z$, $\nu > 0$, compute $Z(\beta)$, both energy probabilities, $\langle H \rangle$ and $\mathrm{Var}\, H$. Check the zero- and high-temperature limits.
2. **A particle in a box.** The energies are $\varepsilon_n = \hbar^2\pi^2 n^2/(2mL^2)$, $n \geq 1$. Show the Gibbs partition sum is finite for $\beta > 0$ by comparison with a geometric series. Give the probabilities and the zero-temperature limit. No closed form for the sum is required.
3. **Why the free particle differs (longer).** In momentum space its Gibbs operator is multiplication by $f(p) = e^{-\beta p^2/(2m)}$. Choose $p_0 > 0$ so $f \geq c > 0$ on $[-p_0, p_0]$. For any finite-rank operator T, find a unit vector supported in that interval with $Tu = 0$ and deduce $\|f(P) - T\| \geq c$ (here $f(P)$ denotes this momentum multiplication). Conclude that this Gibbs operator is not compact and hence not a positive trace-class operator. Use Lecture 1's proved compactness implication, not a divergent spatial trace formula.

> **First hint**
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> Choose more independent interval-supported vectors than the rank of the finite-rank operator. Their images all lie in its finite-dimensional range.

> **Further hint**
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> Take a nontrivial linear dependence of those images and use the same coefficients on the input vectors. Normalise the resulting nonzero kernel vector, then apply the positive lower bound on the multiplier over its support.